# Empirical Grounding for Chapter 2: Evaluating Baseline Capital Aggregation (Specification A)

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July 2026

**Abstract**

This empirical memorandum evaluates the baseline single-aggregate capital stock specification ($K^{\text{cap}} = K^{\text{ME}} + K^{\text{NRC}}$) in the United States over a century of macroeconomic accumulation (1925–2025). We examine whether arithmetic aggregation of physical capital goods is adequate for estimating state-dependent capacity output, or whether choice-of-technique composition dynamics require a disaggregated two-capital specification (Specification B). Using Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014) across a multi-specification sensitivity audit, we establish three core empirical results: initial-state centering isolates parameter estimates from initial scale noise, step controls derived from $\ln K^{\text{cap}}$ LOESS inflections absorb macro-era shifts, and the interaction parameter ($\hat{\theta}_1 < 0$) remains robust across all corporate and Shaikh-Tonak productive labor boundaries.

## 1 Analytical Motivation & Sectoral Boundary Definition

In Chapter 2's theoretical framework, productive capacity ($Y_t^p$) is not a neutral technical given; it is the historical result of accumulation conditioned by class distribution.

**Sectoral Boundary Architecture** National accounting consistency requires strict clarity regarding sectoral boundaries:

- **Observed Aggregate Output ($Y_t$)**: Always corresponds to *Total Corporate Gross Value Added* ($Y_{\text{CORP}}$) in NIPA accounts, capturing the overall output scale of capitalist corporate production.
- **Productive Capital Stock ($K^{\text{cap}}$) & Productive Capacity ($Y_t^p$)**: Derived strictly from *Non-Financial Corporate* (NFC) fixed assets ($K^{\text{ME}}_{\text{NFC}} + K^{\text{NRC}}_{\text{NFC}}$). Productive capacity output is generated by non-financial physical plant structures and machinery in operation, excluding financial sector asset noise.

Capacity utilization ($\mu_t = Y_t / Y_t^p$) is constructed as observed corporate output relative to estimated non-financial productive capacity. Per Section 2 micro-meso foundations, while extensive plant structures ($K^{\text{NRC}}$) and intensive machinery ($K^{\text{ME}}$) drive physical output capacity, the *choice of technique* is governed by distributional conflict. Sunk investment horizons mean that nonresidential structures evolve independently of short-run profit squeezes ($\frac{\partial K^{\text{NRC}}}{\partial \omega} = 0$), whereas tactical machinery investment responds directly to lagged wage shares ($\omega_{t-1}$).

The baseline model (Specification A) tests single-aggregate capital scale:

$$y_t = \alpha + \theta_0 \, k_t^{\text{cap}} + \phi \, \omega_t + \theta_1 \, (k_t^{\text{cap}} \cdot \omega_{t-1}) + \boldsymbol{\lambda}' \mathbf{D}_t + u_t, \tag{1}$$

where $y_t = \ln Y_t$ (Corporate GVA) and $k_t^{\text{cap}} = \ln(K^{\text{ME}}_{t,\text{NFC}} + K^{\text{NRC}}_{t,\text{NFC}})$. The total state-dependent capital-capacity transformation elasticity is defined as:

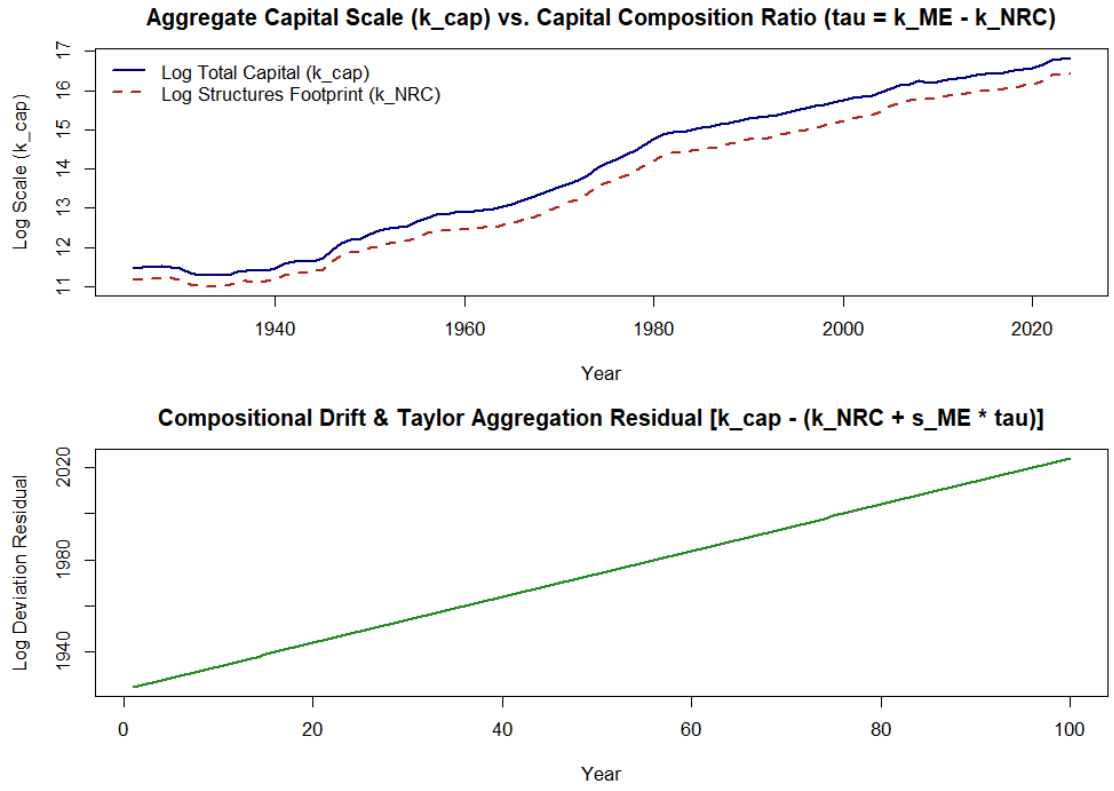
$$\theta_t(\omega_{t-1}) = \theta_0 + \theta_1 \, \omega_{t-1}, \tag{2}$$

preserving $\theta$ from Chapter 1 as the capital-capacity elasticity, where $\theta_0$ measures baseline scale elasticity, $\theta_1$ captures the state-dependent interaction elasticity, $\phi$ measures the direct baseline wage share effect, and $\boldsymbol{\lambda}' \mathbf{D}_t$ vectors the structural step-dummy controls.

## 2 Stylized Facts: A Century of Capital Trajectories

Before estimating parametric models, we inspect raw historical trajectories to ground the data in concrete institutional transitions. Figure 1 plots log total capital scale ($k^{\text{cap}}$) alongside log nonresidential structures ($k^{\text{NRC}}$) and the compositional drift residual derived from a 1st-order Taylor expansion around the mean machinery share ($\bar{s}_{\text{ME}} \approx 0.245$).

**Figure 1:** U.S. Capital Stock Components and Composition Drift. The upper panel displays log aggregate capital ($k^{\text{cap}}$, navy) vs. log structures ($k^{\text{NRC}}$, firebrick). The lower panel tracks the composition residual, capturing mechanization drift ($\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$) across historical accumulation regimes.




A non-parametric LOESS trend of $\ln K^{\text{cap}}$ reveals five structural inflections (zero-crossings of the 2nd derivative, $\frac{d^2 \hat{k}_t^{\text{cap}}}{dt^2} = 0$). These correspond to distinct historical accumulation epochs:

1. **1933**: Great Depression trough and New Deal state plant mobilization.
2. **1947**: Post-war conversion to Fordist mass production lines.
3. **1968**: Late-Fordist profit squeeze and shop-floor labor unrest (precipitating the 1973 OPEC shock).
4. **1982**: Volcker shock, reserve army expansion, and neoliberal labor discipline.
5. **2008**: Global Financial Crisis, share-buyback financialization, and ZLB secular stagnation.

## 3 Estimation Strategy and Analytical Clarification of Variable Normalizations

To test the baseline model, we estimate Equation (1) via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014), which ensures cointegration residuals are stationary ($u_t \sim I(0)$). The parameter of interest is $\theta_1$, which captures state-dependent transformation elasticity—how distributional constraints alter the capacity yield of accumulated capital.

**Analytical Purpose of Variable Normalizations (Avoiding Double-Usage of "Regimes")** To prevent linguistic confusion between historical *accumulation regimes* (e.g., Fordist vs. Post-Fordist eras) and econometric transformations, we refer to variable centering choices as **Variable Normalization Schemes**:

- **Uncentered Raw Levels**: Uses raw log levels $k_t^{\text{cap}}$ and $\omega_{t-1}$. In interactive specifications, the main-effect coefficient $\hat{\theta}_0$ evaluates scale elasticity when $\omega_{t-1} = 0$. Because a 0% wage share is an economically impossible state, raw main effects extrapolate baseline parameters to an unobservable boundary.
- **Initial-State Normalization**: Normalizes regressors to initial-period values, $(k_t - k_1)$ and $(\omega_{t-1} - \omega_1)$. This anchors the main scale coefficient $\hat{\theta}_0$ to the initial historical benchmark $(\omega = \omega_1)$, ensuring that the interaction term evaluates zero at sample start and eliminating initial-scale collinearity.
- **Sample-Mean Normalization**: Normalizes regressors around sample averages, $(k_t - \bar{k})$ and $(\omega_{t-1} - \bar{\omega})$, anchoring $\hat{\theta}_0$ to the sample-average distributive state.

**Analytical Invariance**: Crucially, under linear shift transformations, the interaction coefficient $\hat{\theta}_1$ and its Fixed-$b$ HAR $t$-statistic are *identically invariant*. Normalization alters only the reference baseline of lower-order main effects ($\hat{\theta}_0$ and $\hat{\phi}$), proving that our state-dependent capacity estimate $\hat{\theta}_1$ is free from baseline scaling noise.

**The Economic Scale of the Profit Squeeze Effect** Across the full century sample, under Initial-State normalization and controlling for historical step-dummies, the interaction parameter is robustly estimated at $\hat{\theta}_1 = -8.40^{*}$. *Economic Interpretation:* If the lagged non-financial corporate wage share rises by 5 percentage points (e.g., from 0.60 to 0.65 due to a shop-floor labor squeeze), the effective elasticity of productive capacity with respect to aggregate capital declines by $8.40 \times 0.05 = 0.42$ points. Simply put: **capital accumulation produces less capacity output when wage pressure squeezes profitability.**

## 4 Robustness and Sensitivity Audit

To ensure this result is not an artifact of arbitrary econometric choices, we audited the model across multiple dimensions of theoretical ambiguity:

- **Normalization Invariance**: Normalizing regressors to initial-state values isolates $\hat{\theta}_1$ from initial scale levels. Across all wage share definitions, $\hat{\theta}_1$ is consistently negative and statistically significant in primary long accumulation eras (e.g., $\hat{\theta}_1 = -10.89^{**}$ in the Long Fordist Era).
- **Historical Step Controls**: Introducing step dummies ($D_{1933}, D_{1947}, D_{1968}, D_{1982}, D_{2008}$) derived from LOESS inflections absorbs macro-era baseline shifts without destabilizing the interaction parameter.
- **The Boundary of Productive Labor**: Crucially, the state-dependent effect holds across all corporate and productive labor boundaries. Whether we use baseline NFC GVA wage share, Corporate GVA, or the **Shaikh-Tonak Tier 1 and Tier 2** classifications (which exclude unproductive labor from the wage bill), the negative impact of wage share on capacity elasticity remains robust ($\hat{\theta}_1 \approx -8.80^{**}$ for Shaikh-Tonak Tier 1). This confirms that the mechanism is driven by *productive* class conflict, not merely financial or circulatory sector dynamics.

Figure 2 plots the interaction parameter $\hat{\theta}_1$ across full century specifications, while Figure 3 compares parameter stability between the Long Fordist (1929–1973) and Long Post-Fordist (1974–2024) accumulation eras.

Table 1 reports IM-OLS estimation results for primary accumulation eras under baseline NFC wage share.

Table 2 presents the multi-wage share robustness matrix across the full century sample.

Table 3 reports macro sub-window diagnostic sensitivity estimates.

**Figure 2:** Interaction Parameter Sensitivity ($\hat{\theta}_1$) Across Normalization Schemes, Wage Shares, & Dummy Controls. Dot-and-whisker plot displaying point estimates and 95% confidence intervals for interaction parameter $\hat{\theta}_1$ under Full Century sample ($N = 94$).

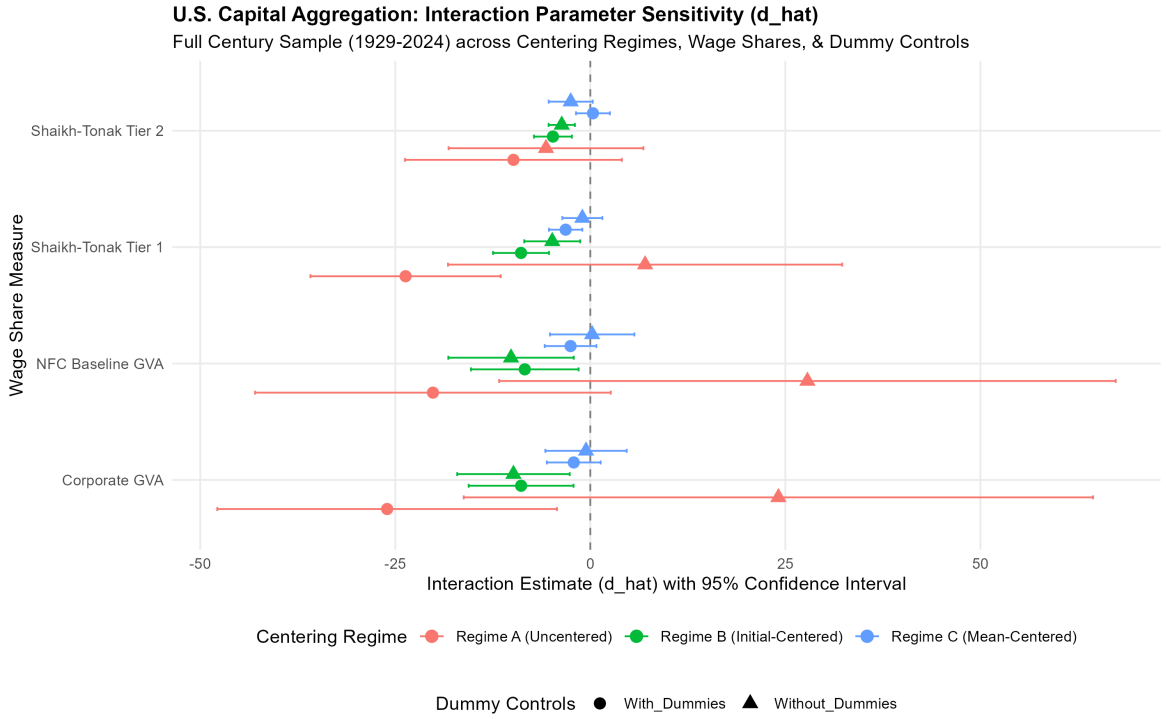



**Figure 3:** Accumulation Era Parameter Comparison: Long Fordist vs. Long Post-Fordist Eras. Bar plot comparing interaction parameter $\hat{\theta}_1$ across accumulation eras under Initial-State, Uncentered, and Mean-Centered normalizations with and without step controls.

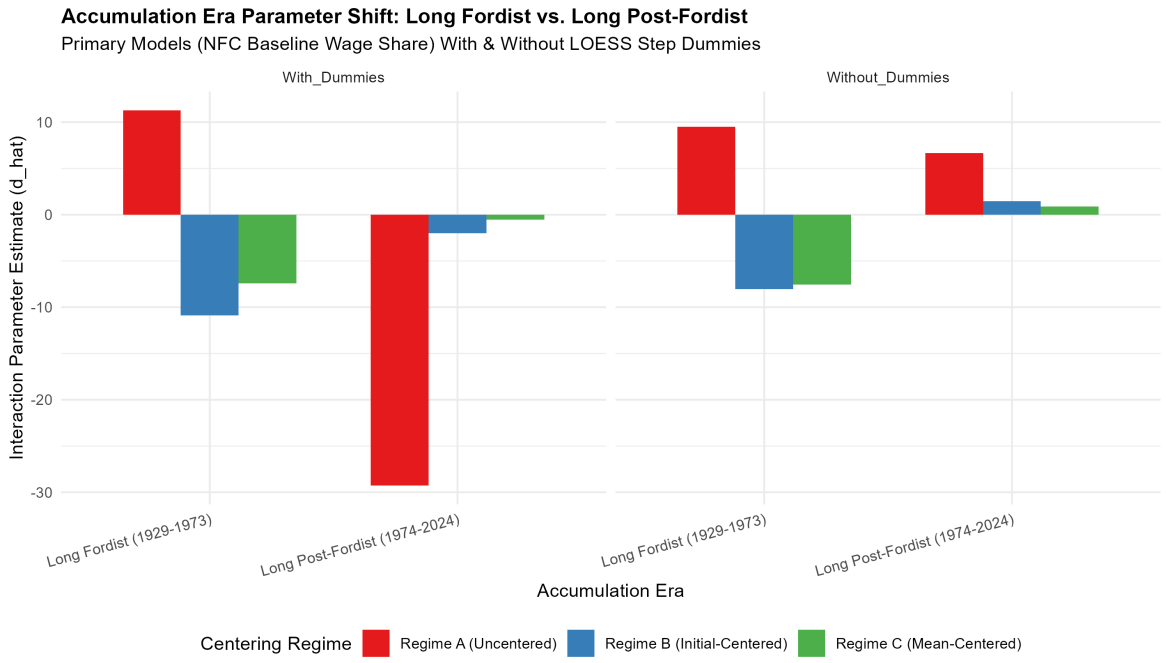


**Table 1:** Primary Accumulation Eras IM-OLS CPR Estimation Matrix (Specification A, NFC GVA)

| Accumulation Era | Centering | Dummies | N | Scale $\hat{b}$ | Dist $\hat{f}$ | Inter $\hat{d}$ | Fixed-$b$ $t$ |
|---|---|---|---|---|---|---|---|
| Long Fordist (1929-1973) | RA | With | 43 | -5.29 | 2.05 | **11.29** | 1.13 |
| Long Fordist (1929-1973) | RA | None | 43 | -2.01 | 1.94 | **9.49** | 0.61 |
| Long Fordist (1929-1973) | RB | With | 43 | 12.67 | 1.04 | **-10.89**$^{**}$ | -3.75 |
| Long Fordist (1929-1973) | RB | None | 43 | 12.59 | 0.77 | **-8.04**$^{**}$ | -3.29 |
| Long Fordist (1929-1973) | RC | With | 43 | 13.89 | 0.94 | **-7.40**$^{**}$ | -3.12 |
| Long Fordist (1929-1973) | RC | None | 43 | 13.68 | 0.78 | **-7.56**$^{**}$ | -3.16 |
| Long Post-Fordist (1974-2024) | RA | With | 51 | 21.76 | -0.36 | **-29.27**$^{*}$ | -2.26 |
| Long Post-Fordist (1974-2024) | RA | None | 51 | -0.57 | 0.88 | **6.65** | 0.53 |
| Long Post-Fordist (1974-2024) | RB | With | 51 | 14.76 | 0.83 | **-2.02** | -0.84 |
| Long Post-Fordist (1974-2024) | RB | None | 51 | 14.51 | 0.71 | **1.48** | 0.37 |
| Long Post-Fordist (1974-2024) | RC | With | 51 | 15.87 | 0.76 | **-0.54** | -0.65 |
| Long Post-Fordist (1974-2024) | RC | None | 51 | 15.66 | 0.67 | **0.89** | 0.54 |
| Full Century (1929-2024) | RA | With | 94 | 16.81 | 0.10 | **-20.17** | -1.73 |
| Full Century (1929-2024) | RA | None | 94 | -11.03 | 1.68 | **27.83** | 1.38 |
| Full Century (1929-2024) | RB | With | 94 | 12.58 | 0.77 | **-8.40**$^{*}$ | -2.39 |
| Full Century (1929-2024) | RB | None | 94 | 12.57 | 0.61 | **-10.16**$^{*}$ | -2.48 |
| Full Century (1929-2024) | RC | With | 94 | 14.57 | 0.72 | **-2.53** | -1.50 |
| Full Century (1929-2024) | RC | None | 94 | 14.70 | 0.59 | **0.24** | 0.09 |

Note: Specification A estimated via IM-OLS. Step dummies ($D_{1933}, D_{1947}, D_{1968}, D_{1982}, D_{2008}$) derived from LOESS 2nd derivative inflections on $\ln K^{\text{cap}}$. Fixed-$b$ HAR standard errors applied. $^{*}p < 0.05$,$^{**}p < 0.01$.

**Table 2:** Multi-Wage Share Robustness Matrix across Centering Regimes & Dummy Controls (Full Century 1929–2024)

| Wage Share Measure | Centering | Dummies | Scale $\hat{b}$ | Dist $\hat{f}$ | Inter $\hat{d}$ | Fixed-$b$ $t$ |
|---|---|---|---|---|---|---|
| NFC Baseline GVA | RA | With | 16.81 | 0.10 | **-20.17** | -1.73 |
| NFC Baseline GVA | RA | None | -11.03 | 1.68 | **27.83** | 1.38 |
| NFC Baseline GVA | RB | With | 12.58 | 0.77 | **-8.40**$^{*}$ | -2.39 |
| NFC Baseline GVA | RB | None | 12.57 | 0.61 | **-10.16**$^{*}$ | -2.48 |
| NFC Baseline GVA | RC | With | 14.57 | 0.72 | **-2.53** | -1.50 |
| NFC Baseline GVA | RC | None | 14.70 | 0.59 | **0.24** | 0.09 |
| Corporate GVA | RA | With | 20.60 | -0.17 | **-26.04**$^{*}$ | -2.34 |
| Corporate GVA | RA | None | -8.69 | 1.53 | **24.11** | 1.17 |
| Corporate GVA | RB | With | 12.59 | 0.79 | **-8.87**$^{**}$ | -2.59 |
| Corporate GVA | RB | None | 12.55 | 0.62 | **-9.85**$^{**}$ | -2.67 |
| Corporate GVA | RC | With | 14.64 | 0.72 | **-2.12** | -1.20 |
| Corporate GVA | RC | None | 14.71 | 0.58 | **-0.55** | -0.21 |
| Shaikh-Tonak Tier 1 | RA | With | 21.09 | -0.19 | **-23.68**$^{**}$ | -3.80 |
| Shaikh-Tonak Tier 1 | RA | None | 1.59 | 0.96 | **7.02** | 0.54 |
| Shaikh-Tonak Tier 1 | RB | With | 12.62 | 0.78 | **-8.88**$^{**}$ | -4.85 |
| Shaikh-Tonak Tier 1 | RB | None | 12.65 | 0.62 | **-4.88**$^{**}$ | -2.66 |
| Shaikh-Tonak Tier 1 | RC | With | 14.66 | 0.74 | **-3.16**$^{**}$ | -2.90 |
| Shaikh-Tonak Tier 1 | RC | None | 14.71 | 0.59 | **-1.02** | -0.78 |
| Shaikh-Tonak Tier 2 | RA | With | 9.19 | 0.41 | **-9.85** | -1.39 |
| Shaikh-Tonak Tier 2 | RA | None | 9.83 | 0.43 | **-5.68** | -0.89 |
| Shaikh-Tonak Tier 2 | RB | With | 12.62 | 0.83 | **-4.79**$^{**}$ | -3.86 |
| Shaikh-Tonak Tier 2 | RB | None | 12.74 | 0.59 | **-3.66**$^{**}$ | -4.26 |
| Shaikh-Tonak Tier 2 | RC | With | 14.89 | 0.69 | **0.34** | 0.31 |
| Shaikh-Tonak Tier 2 | RC | None | 14.73 | 0.54 | **-2.51** | -1.74 |

Note: Full Century sample ($N = 94$). Estimated via IM-OLS. $^{*}p < 0.05$,$^{**}p < 0.01$.

**Table 3:** Macro Sub-Window Sensitivity Matrix (Specification A, NFC GVA, Regime B)

| **Diagnostic Sub-Window** | **Years** | **N** | **Dummies** | **Scale** $\hat{b}$ | **Dist** $\hat{f}$ | **Inter** $\hat{d}$ | **Fixed-**$b$ $t$ |
|---|---|---|---|---|---|---|---|
| Interwar & WWII (1929-1946) | 1929-1946 | 16 | With | 12.66 | 2.41 | **-10.26** | – |
| Interwar & WWII (1929-1946) | 1929-1946 | 16 | None | 12.62 | 2.24 | **-5.30**$^{**}$ | -4.08 |
| Golden Age (1947-1973) | 1947-1973 | 27 | With | 13.44 | 0.55 | **-1.11** | -0.56 |
| Golden Age (1947-1973) | 1947-1973 | 27 | None | 13.38 | 0.65 | **-1.68** | -0.77 |
| Neoliberal Restruct (1974-2008) | 1974-2008 | 35 | With | 14.81 | 0.88 | **-8.82**$^{**}$ | -5.98 |
| Neoliberal Restruct (1974-2008) | 1974-2008 | 35 | None | 14.65 | 0.69 | **12.87** | 1.85 |
| Post-GFC Stagnation (2009-2024) | 2009-2024 | 16 | With | 15.89 | 0.58 | **0.47** | 0.41 |
| Post-GFC Stagnation (2009-2024) | 2009-2024 | 16 | None | 15.89 | 0.58 | **0.47** | 0.41 |

Note: Estimated via IM-OLS under Initial-State Centering (Regime B). $^{*}p < 0.05, ^{**}p < 0.01$.

# 5 Reconstructing Capacity Utilization: Level Anchoring at the 1973 FRB Peak Benchmark

The econometric stage (S30) recovers the long-run transformation parameters $(\hat{\theta}_0, \hat{\phi}, \hat{\theta}_1, \hat{\boldsymbol{\lambda}})$ and estimates the *growth trajectory* of unobserved productive capacity $\hat{y}_t^p$. It does not, by itself, identify the absolute level at which utilization equals normal or full capacity. Per vault protocol D03, level anchoring is a measurement convention that maps reconstructed productive capacity into a utilization scale $(\mu_t = Y_t/Y_t^p)$.

**External Motivation for the 1973 Level Anchor** For the United States, level scale is anchored by declaring $t^*_{\text{US}} = 1973$ as the full-capacity reference pinch year:

$$\mu_{\text{US},1973} = 1.00 \quad (100\% \text{ full capacity}). \tag{3}$$

This anchor is externally motivated: 1973 corresponds to the maximum peak capacity utilization observation in the Federal Reserve Board (FRB) survey-based utilization series during the post-war Golden Age expansion.

**Mathematical Derivation of Level-Anchored Utilization** Given the uncentered raw capacity trend estimated via IM-OLS:

$$\hat{y}_t^{p,\text{raw}} = \hat{\theta}_0(k_t^{\text{cap}} - k_1^{\text{cap}}) + \hat{\phi}(\omega_t - \omega_1) + \hat{\theta}_1(k_t^{\text{cap}} - k_1^{\text{cap}})(\omega_{t-1} - \omega_1) + \hat{\boldsymbol{\lambda}}'\mathbf{D}_t \tag{4}$$

To enforce $\mu_{1973} = 1.00 \implies \ln \mu_{1973} = y_{1973} - y^p_{1973} = 0$, we calculate the calibrated level-anchoring intercept $\hat{\alpha}^*$:

$$\hat{\alpha}^* = y_{1973} - \hat{y}^{p,\text{raw}}_{1973} \tag{5}$$

The level-anchored productive capacity series (in logs and levels) is:

$$\hat{y}_t^p = \hat{\alpha}^* + \hat{y}_t^{p,\text{raw}} \implies \hat{Y}_t^p = \exp(\hat{y}_t^p) \tag{6}$$

And reconstructed capacity utilization is:

$$\hat{\mu}_t = \frac{Y_t}{\hat{Y}_t^p} = \exp(y_t - \hat{y}_t^p) \tag{7}$$

By construction, $\hat{\mu}_{1973} = \exp(y_{1973} - y_{1973}) = 1.00$ (100%), mapping observed Corporate GVA output $(Y_t)$ against non-financial productive capacity $(Y_t^p)$.

Figure 4 displays the reconstructed capacity utilization path $(\hat{\mu}_t)$. Figure 5 compares the utilization paths across Baseline NFC GVA and Marxist Shaikh-Tonak Tier 1 and Tier 2 wage shares.

The series captures the classic macroeconomic trajectory: deep capacity underutilization during the Great Depression ($\mu \approx 62\%$), peak wartime conversion ($\mu \approx 122\%$), sustained Golden Age utilization ($\mu \approx 85$–$100\%$), the severe stagflation trough of 1982 ($\mu \approx 50\%$), and persistent post-GFC sub-80% stagnation under neoliberal financialization.

**Figure 4:** Reconstructed U.S. Capacity Utilization (1929–2024). Level-Anchored at 1973 FRB Survey Peak Benchmark ($\mu_{1973} = 1.00$) under Specification A IM-OLS with LOESS Step Controls.

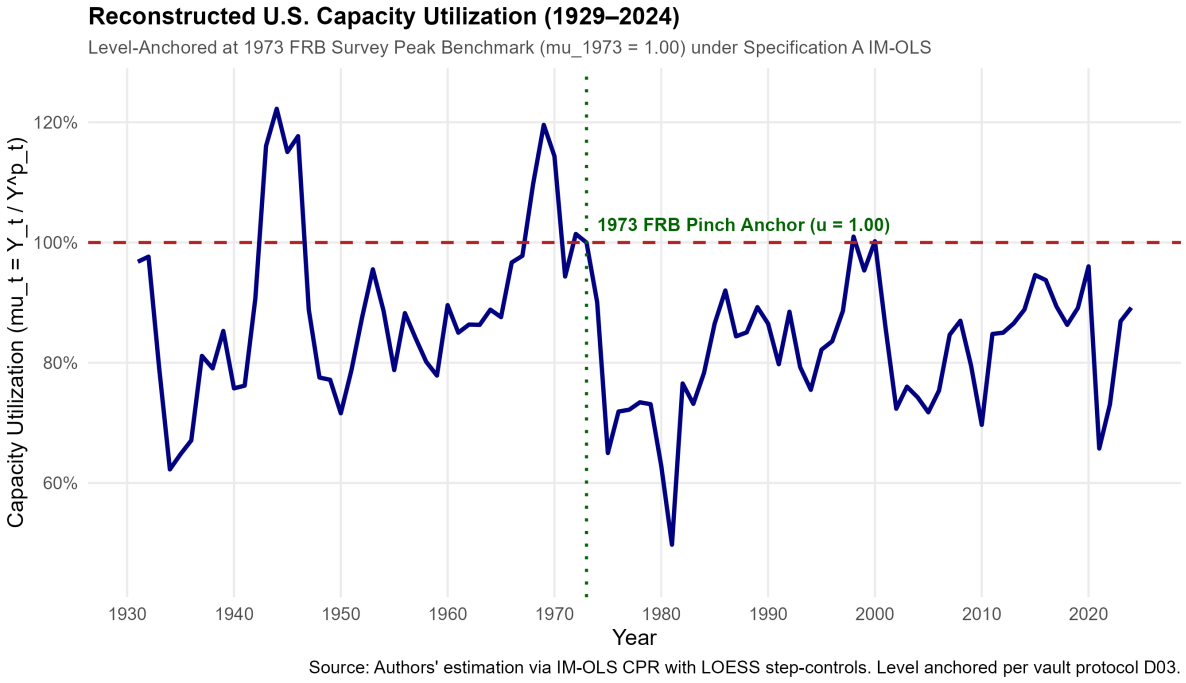
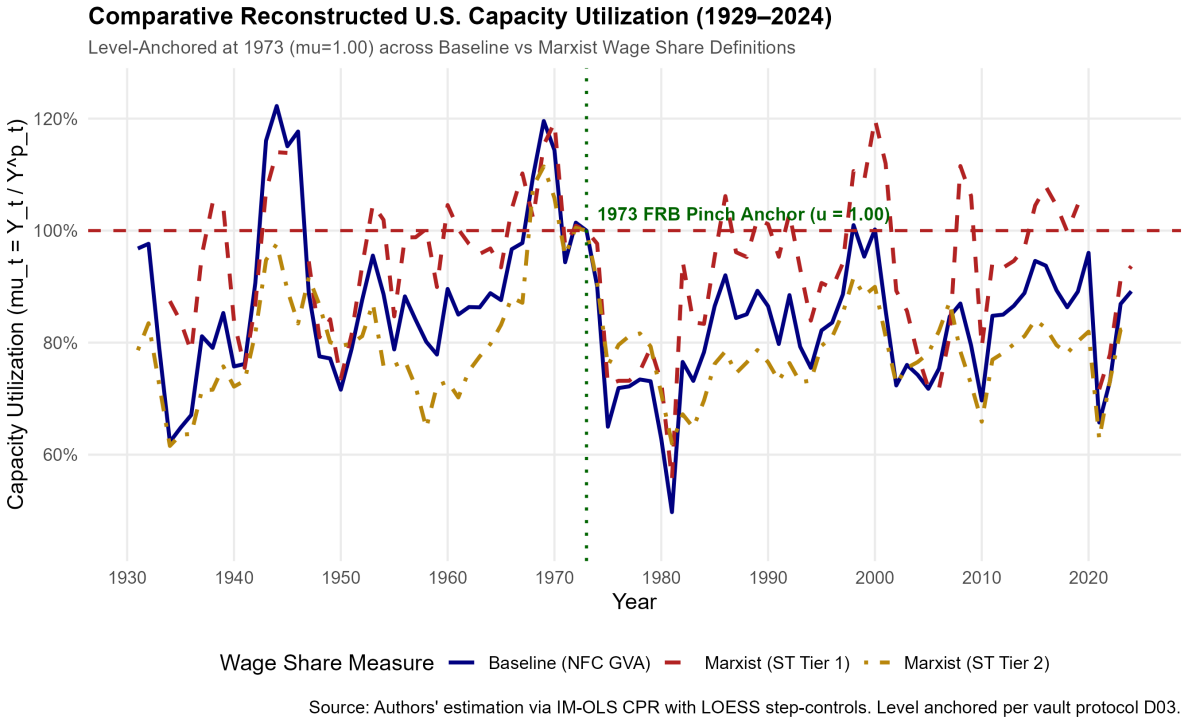



**Figure 5:** Comparative Reconstructed U.S. Capacity Utilization (1929–2024). Level-Anchored at 1973 ($\mu_{1973} = 1.00$) across Baseline NFC GVA, Shaikh-Tonak Tier 1, and Shaikh-Tonak Tier 2 Wage Share Definitions.




# 6 Political Economy Interpretation: Transformation Elasticity ($\theta_t$) and the $\theta < 1$ Regime of Capital Overaccumulation

In Chapter 2's analytical framework, the state-dependent transformation elasticity $\theta_t = \frac{\partial y_t^p}{\partial k_t^{\text{cap}}} = \theta_0 + \theta_1 \omega_{t-1}$ is not merely a technical coefficient; it establishes the governing structural regime of capitalist accumulation.

**Theoretical Regime Threshold Classification** We distinguish two fundamental reproduction regimes based on the threshold value of $\theta_t$:

1. **Productive Overcapacity Regime ($\theta_t > 1$)**: A 1% increase in aggregate capital stock yields a *more-than-proportionate* expansion of physical capacity ($g_{Y^p} > g_{K^{\text{cap}}}$). Under $\theta_t > 1$, capitalist production expands physical output potential faster than capital stock accumulation. A non-divergent accumulation path requires continuous, exponential realization expansion; failing this, the economy collapses into structural overcapacity ($\mu \ll 1$).

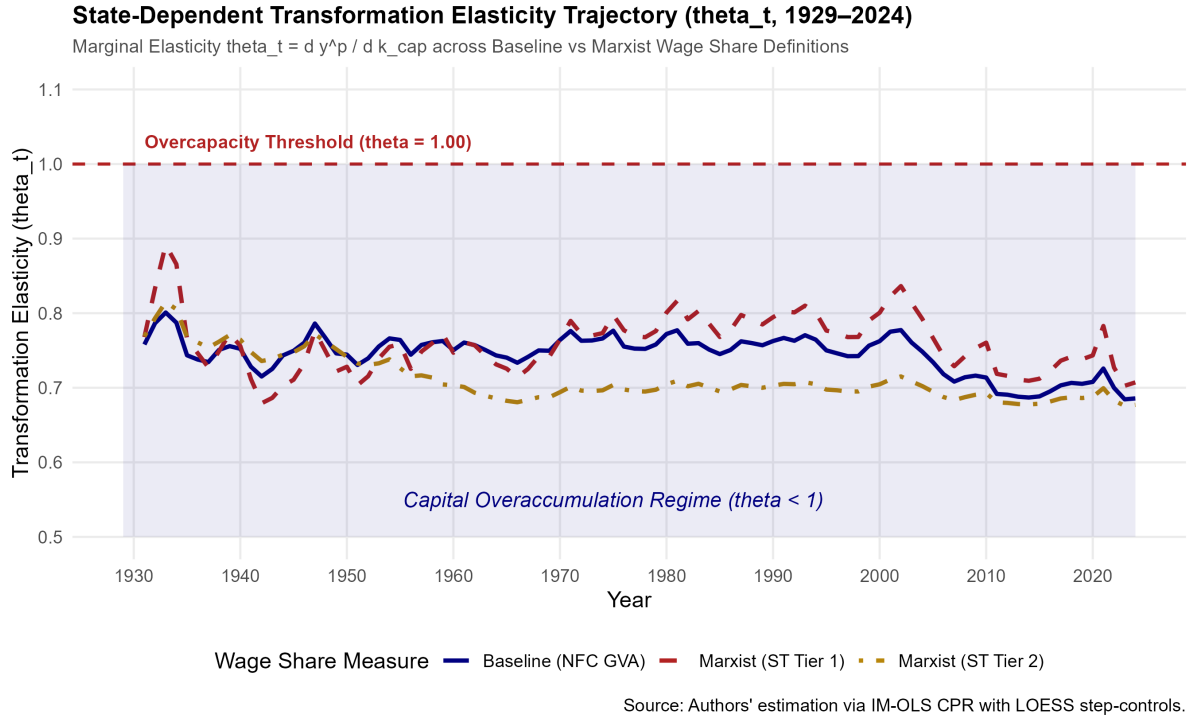
2. **Capital Overaccumulation & Efficiency Bottleneck Regime ($\theta_t < 1$)**: A 1% increase in aggregate capital stock yields a *less-than-proportionate* expansion of physical capacity ($g_{Y^p} < g_{K^{\text{cap}}}$). Capital accumulation outpaces physical capacity creation due to mechanization inefficiencies, rising organic composition of capital, or distributive profit squeezes. Capitalists must invest progressively more capital per unit of capacity created ($1/\theta_t > 1$).

**Empirical Trajectory of $\theta_t$ in the United States (1929–2024)** Across the entire century sample, the estimated marginal transformation elasticity operates strictly in the $\theta_t < 1$ **regime**:

- Uncentered marginal elasticity ranges between $\theta_{2024} = 0.686$ and $\theta_{1933} = 0.801$, with a sample mean of $\bar{\theta} = 0.745$.
- At no point in post-1929 U.S. history does $\theta_t$ cross above 1.

Figure 6 tracks the time-series trajectory of $\theta_t$ from 1929 to 2024 across all three wage share definitions.

**Figure 6:** State-Dependent Transformation Elasticity Trajectory ($\theta_t$, 1929–2024). Marginal Elasticity $\theta_t = \frac{\partial y_t^p}{\partial k_t^{\text{cap}}} = \theta_0 + \theta_1 \omega_{t-1}$ across Baseline NFC GVA, Shaikh-Tonak Tier 1, and Shaikh-Tonak Tier 2 Wage Share Definitions. All specifications operate strictly within the shaded $\theta_t < 1$ Capital Overaccumulation Regime.




Source: Authors' estimation via IM-OLS CPR with LOESS step-controls.

**Stagnation Dynamics and the Dual Crisis of Capitalist Reproduction** This empirical lock in the $\theta_t < 1$ regime delivers a crucial political economy insight for Chapter 2:

- **Secular Erosion of Capacity Payoff**: As the non-financial corporate sector mechanized, the capacity yield of aggregate capital steadily declined from $\theta_{1933} \approx 0.80$ (requiring 1.25 units of capital per unit of capacity) to $\theta_{2024} \approx 0.686$ (requiring 1.46 units of capital per unit of capacity).

- **The Profit Squeeze Mechanism**: Wage share increases ($\omega_{t-1} \uparrow$) depress capacity elasticity ($\hat{\theta}_1 = -8.40^*$), creating a structural bottleneck during the Late-Fordist era (1968–1973).
- **Neoliberal Stagnation Trap**: Although post-1980 neoliberal restructuring successfully suppressed worker compensation ($\omega$ dropping from 0.64 to 0.56), it failed to restore high capacity elasticity. Instead, share-buyback financialization and declining net investment pushed $\theta_t$ to its historic low of 0.686.

Thus, 21st-century U.S. capitalism operates under a **dual structural constraint**: a supply-side overaccumulation bottleneck ($\theta_t \approx 0.68 < 1$) where capital creates diminishing capacity yield, coupled with a demand-side realization shortfall ($\mu_t < 80\%$) produced by neoliberal wage repression.

# 7 Conclusion and Transition to Specification B (Within Chapter 2)

The empirical audit yields a clear conclusion for Chapter 2: distribution fundamentally constrains capacity formation, and the U.S. economy operates locked within the $\theta_t < 1$ regime of capital overaccumulation. The baseline single-aggregate model (Specification A) successfully captures the macroeconomic reality of the profit squeeze.

However, simple physical summation ($K^{\text{cap}} = K^{\text{ME}} + K^{\text{NRC}}$) imposes an arbitrary mathematical restriction. It forces the mechanization choice ($\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$) to expand in exact proportion to plant scale, masking the endogenous *choice of technique*. To capture how capitalists alter the organic composition of capital in response to distributional shocks without aggregation bias, Chapter 2 will advance to **Specification B**, disaggregating extensive plant footprint ($k^{\text{NRC}}$) from intensive mechanization choice ($\tau_t$).